

# Yufei Chen

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(Last updated: January 2026)  
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## OVERVIEW

My research focuses on the intersection of cybersecurity and artificial intelligence (AI). Currently, I am exploring new attack surfaces introduced by AI components and investigating how AI can be utilized to enhance existing security measures.

My (co-)first-author papers have been published in top security or machine learning venues: **USENIX Security** ( $\times 2$ ), **NDSS** ( $\times 1$ ), **TDSC** ( $\times 1$ ), and **NeurIPS** ( $\times 1$ ). For the latest updates on my publications, please visit [my Google Scholar profile](#) (**Google scholar citations: 1400+**).

## EDUCATION

- 09/2019 – 06/2024 **City University of Hong Kong**, Hong Kong SAR, China  
Ph.D. in Computer Science (Dual Ph.D. Programme)  
Supervisor: Cong Wang
- 09/2016 – 12/2023 **Xi'an Jiaotong University**, Xi'an, China  
Ph.D. in Control Science and Engineering (Dual Ph.D. Programme)  
Supervisor: Chao Shen
- 02/2014 – 06/2016 **Xi'an Jiaotong University**, Xi'an, China  
B.B.A. in Business Administration
- 09/2012 – 06/2016 **Xi'an Jiaotong University**, Xi'an, China  
B.Eng. in Electrical Engineering

## WORKING EXPERIENCES

- 01/2026 – PRESENT **City University of Hong Kong**, Hong Kong SAR, China  
Research Assistant Professor  
Department of Computer Science
- 06/2024 – 01/2026 **City University of Hong Kong**, Hong Kong SAR, China  
Postdoc  
Department of Computer Science  
Supervisor: Cong Wang
- 03/2018 – 11/2018 **Qihoo 360**, Beijing, China  
Research Intern  
Research Center for Intelligence Security  
Supervisor: Kang Li

## PUBLICATIONS

(\*: co-first authors)

### CONFERENCE PAPERS

- C15. Longxiang Wang\*, Lei Xu\*, **Yufei Chen**, Ying Zou, and Cong Wang. “ALERT: Machine Learning-Enhanced Risk Estimation for Databases Supporting Encrypted Queries,” in *Proceedings of 34th USENIX Security Symposium (USENIX Security)*, 2025. (Help with research idea refinement and paper editing.)
- C14. Sen Peng, **Yufei Chen**, Cong Wang, and Xiaohua Jia. “Intellectual Property Protection of Diffusion Models via the Watermark Diffusion Process,” in *Proceedings of 25th International Web Information Systems Engineering Conference (WISE)*, 2024. (Help with research idea refinement and paper editing.)
- C13. Xuanqi Gao, Juan Zhai, Shiqing Ma, Chao Shen, **Yufei Chen**, and Shiwei Wang. “CILATE: Towards Fairer Class-based Incremental Learning by Dataset and Training Refinement,” in *Proceedings of 32nd ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA)*, 2023. (Help with research idea refinement and paper editing.)
- C12. **Yufei Chen**, Chao Shen, Yun Shen, Cong Wang, and Yang Zhang. “Amplifying Membership Exposure via Data Poisoning,” in *Proceedings of 36th Conference on Neural Information Processing Systems (NeurIPS)*, 2022. (Lead the research project, from algorithm design, implementation, and paper writing.)
- C11. **Yufei Chen**, Chao Shen, Cong Wang, and Yang Zhang. “Teacher Model Fingerprinting Attacks Against Transfer Learning,” in *Proceedings of 31st USENIX Security Symposium (USENIX Security)*, 2022. (Lead the research project, from algorithm design, implementation, and paper writing.)
- C10. Xuanqi Gao, Juan Zhai, Shiqing Ma, Chao Shen, **Yufei Chen**, and Qian Wang. “FairNeuron: Improving Deep Neural Network Fairness with Adversary Games on Selective Neurons,” in *Proceedings of 44th International Conference on Software Engineering (ICSE)*, 2022. (Help with research idea refinement and paper editing.)
- C9. Junhao Zhou\*, **Yufei Chen\***, Chao Shen, and Yang Zhang. “Property Inference Attacks Against GANs,” in *Proceedings of the Network and Distributed System Security Symposium (NDSS)*, 2022. (Lead evaluation design and paper writing.)
- C8. Qixue Xiao\*, **Yufei Chen\***, Chao Shen, Yu Chen, and Kang Li. “Seeing is Not Believing: Camouflage Attacks on Image Scaling Algorithms,” in *Proceedings of 28th USENIX Security Symposium (USENIX Security)*, 2019. (Lead attack algorithm design, implementation, and paper writing.)

- C7. Chao Shen, Qi Lv, Zhao Wang, **Yufei Chen**, and Xiaohong Guan. “Hand-Interactive Behavior Analysis for User Authentication Systems with Wrist-Worn Devices,” in *Proceedings of 2018 International Conference on Information, Cybernetics, and Computational Social Systems (ICCSS)*, 2018. Best Paper Award 🏆
- C6. Xiaozhi Liu, Chao Shen, and **Yufei Chen**. “Multi-source Interactive Behavior Analysis for Continuous User Authentication on Smartphones,” in *Proceedings of 2018 Chinese Conference on Biometric Recognition (CCBR)*, 2018.
- C5. Zhao Wang, Chao Shen, and **Yufei Chen**. “Handwaving Authentication: Unlocking Your Smartwatch Through Handwaving Biometrics,” in *Proceedings of 2017 Chinese Conference on Biometric Recognition (CCBR)*, 2017.
- C4. Chao Shen, Ziqiang Ren, **Yufei Chen**, and Zhao Wang. “On Using Wearable Devices to Steal Your Passwords: A Fuzzy Inference Approach,” in *Proceedings of 2017 International Symposium on Cyberspace Safety and Security*, 2017.
- C3. Zhanpei Jia, Chao Shen, Xiao Yi, **Yufei Chen**, Tianwen Yu, and Xiaohong Guan. “Big-data analysis of multi-source logs for anomaly detection on network-based system,” in *Proceedings of 13th IEEE Conference on Automation Science and Engineering (CASE)*, 2017.
- C2. **Yufei Chen**, Chao Shen, Zhao Wang, and Tianwen Yu. “Modeling Interactive Sensor-Behavior with Smartphones for Implicit and Active User Authentication,” in *Proceedings of 2017 IEEE International Conference on Identity, Security and Behavior Analysis (ISBA)*, 2017.
- C1. Chao Shen, **Yufei Chen**, and Gengshan Yang. “On Motion-Sensor Behavior Analysis for Human-Activity Recognition via Smartphones,” in *Proceedings of 2016 IEEE International Conference on Identity, Security and Behavior Analysis (ISBA)*, 2016.
- J11. Sen Peng, **Yufei Chen**, Jie Xu, Zizhuo Chen, Cong Wang, and Xiaohua Jia. “Intellectual Property Protection of DNN Models,” *World Wide Web*, vol. 26, pp. 1877-1911, 2023.
- J10. Kaidi Jin, Tianwei Zhang, Chao Shen, **Yufei Chen**, Ming Fan, Chenhao Lin, and Ting Liu. “Can We Mitigate Backdoor Attack Using Adversarial Detection Methods?,” *IEEE Transactions on Dependable and Secure Computing (TDSC)*, vol. 20, no. 4, pp. 2867-2881, 2023.
- J9. Xinyan Li, **Yufei Chen**, Cong Wang, and Chao Shen. “When Deep Learning Meets Differential Privacy: Privacy, Security, and More,” *IEEE Network*, vol. 35, no. 6, pp. 148-155, 2021.
- J8. **Yufei Chen**, Chao Shen, Cong Wang, Qixue Xiao, Kang Li, and Yu Chen. “Scaling Camouflage: Content Disguising Attack Against Computer Vision Applications,” *IEEE Transactions on Dependable and Secure Computing (TDSC)*, vol. 18, no. 5, pp. 2017-2028, 2021.

- J7. Chao Shen, Zhao Wang, Chengxiang Si, **Yufei Chen**, and Xiaojie Su. “Waving Gesture Analysis for User Authentication in the Mobile Environment,” *IEEE Network*, vol. 34, no. 2, pp. 57-63, 2020.
- J6. **Yufei Chen**, Chao Shen, Qian Wang, Qi Li, Cong Wang, Shouling Ji, Kang Li, and Xiaohong Guan. “Security and Privacy Risks in Artificial Intelligence Systems (人工智能系统安全与隐私风险),” *Journal of Computer Research and Development*, vol. 56, no. 10, pp. 2135-2150, 2019. *(in Chinese)*
- J5. Chao Shen, **Yufei Chen**, Yao Liu, and Xiaohong Guan. “Adaptive Human-Machine Interactive Behavior Analysis With Wrist-Worn Devices for Password Inference,” *IEEE Transactions on Neural Networks and Learning Systems (TNNLS)*, vol. 29, no. 12, pp. 6292-6302, 2018.
- J4. Chao Shen, **Yufei Chen**, and Xiaohong Guan. “Performance Evaluation of Implicit Smartphones Authentication via Sensor-Behavior Analysis,” *Information Sciences*, vol. 430-431, pp. 538-553, 2018.
- J3. Chao Shen, **Yufei Chen**, Xiaohong Guan, and Roy Maxion. “Pattern-Growth based Mining Mouse-Interaction Behavior for an Active User Authentication System,” *IEEE Transactions on Dependable and Secure Computing (TDSC)*, vol. 17, no. 2, pp. 335-349, 2018.
- J2. Chao Shen, **Yufei Chen**, Gengshan Yang, and Xiaohong Guan. “Toward Hand-Dominated Activity Recognition Systems With Wristband-Interaction Behavior Analysis,” *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, vol. 50, no. 7, pp. 2501-2511, 2018.
- J1. Chao Shen, Yuanxun Li, **Yufei Chen**, Xiaohong Guan, and Roy Maxion. “Performance Analysis of Multi-Motion Sensor Behavior for Active Smartphone Authentication,” *IEEE Transactions on Information Forensics and Security (TIFS)*, vol. 13, no. 1, 2018.

## TEACHING

|             |                                                                                  |
|-------------|----------------------------------------------------------------------------------|
| 2026 Spring | CS5293 (Topics on Information Security), PG, <b>Tutor</b> , CityUHK(DG)          |
| 2025 Fall   | CS6535 (Guided Study in Artificial Intelligence), PG, <b>Tutor</b> , CityUHK(DG) |
| 2025 Spring | CS6290 (Privacy-enhancing Technologies), PG, <b>Tutor</b> , CityUHK(DG)          |
| 2025 Spring | CS2310 (Computer Programming), UG, <b>Tutor</b> , CityUHK                        |
| 2024 Fall   | CS2311 (Computer Programming), UG, <b>Tutor</b> , CityUHK                        |

**Courses can teach:** I am prepared to teach a broad range of undergraduate and graduate courses in computer science, including computer programming, computer security, networking, data mining, and trustworthy AI. My teaching philosophy emphasizes bridging fundamental principles with real-world applications, fostering analytical thinking, and promoting responsible and ethical use of artificial intelligence technologies.

## AWARDS AND HONORS

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|---------|-------------------------------------|
| 11/2025 | NeurIPS 2025 Top Reviewer (Top 8%)  |
| 11/2024 | NeurIPS 2024 Top Reviewer (Top 10%) |

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| 07/2022     | USENIX Security 2022 Student Grant                                                |
| 11/2021     | Outstanding Presenter Award of the 14th USTC PhD Student Workshop                 |
| 03/2021     | The Shaanxi Provincial Natural Science Award (First Class, 4/4)                   |
| 12/2020     | <b>National Scholarship for Ph.D. Student (Top 0.2%)</b>                          |
| 06/2020     | Outstanding Graduate Student of XJTU                                              |
| 10/2019     | Cyber Security Scholarship (awarded annually to 100 graduate students nationwide) |
| 12/2018     | <b>National Scholarship for Ph.D. Student (Top 0.2%)</b>                          |
| 08/2018     | IEEE ICCSS Best Paper Award                                                       |
| 12/2017     | <b>National Scholarship for Graduate Student (Top 0.2%)</b>                       |
| 12/2015     | <b>National Scholarship for Undergraduate Student (Top 0.2%)</b>                  |
| 12/2013     | National Endeavor Scholarship (Top 3%)                                            |
| 2012 – 2016 | Outstanding Undergraduate Student of XJTU                                         |

## SERVICES

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| PC MEMBER<br>REVIEWER | NeurIPS 2025, 2024 (Top reviewer $\times 2$ 🏆)<br>ICLR 2026, 2025<br>ICML 2025<br>AAAI 2025<br>AISTATS 2026, 2025<br>LAMPS-CCS 2025<br>MSN 2025, 2024<br>SecTL-AsiaCCS 2025, 2024, 2023                                                                                                                                                                                                                                                                                                                    |
| EXTERNAL<br>REVIEWER  | USENIX Security 2024, 2023<br>NDSS 2022<br>ACM CCS 2023<br>ESORICS 2024<br>RAID 2022, 2021<br>ACSAC 2022<br>ACNS 2022<br>AsiaCCS 2021<br>ICICS 2021                                                                                                                                                                                                                                                                                                                                                        |
| JOURNAL<br>REVIEWER   | IEEE Transactions on Dependable and Secure Computing<br>IEEE Transactions on Information Forensics and Security<br>IEEE Transactions on Mobile Computing<br>ACM Transactions on Privacy and Security<br>IEEE Transactions on Neural Networks and Learning Systems<br>IEEE Internet of Things Journal<br>IEEE Transactions on Cybernetics<br>ACM Transactions on Sensor Networks<br>Transactions on Machine Learning Research<br>IEEE Transactions on Automation Science and Engineering<br>Neural Networks |

Computers & Security  
Information Sciences